Lecture 5

Mathematical Problems

Example: A body of mass 0.5 kg is suspended from a spring of negligible mass and it stretches the spring by 0.07 m. For a displacement of 0.03 m it has a downward velocity of 0.4 m/sec. Calculate (a) the time period, (b) the frequency and (c) the amplitude of vibration of the spring.

Solution:

From F = ky, the force constant of the spring

$$k = \frac{F}{y} = \frac{0.5 \times 9.8}{0.07}$$

(a) The time period

$T=2\pi\sqrt{\frac{m}{k}}$

$T=2 \times 3.14 \times \sqrt{\frac{0.5\times0.07}{0.5\times9.8}}$= 0.5307 sec.

(b) The frequency

$n = \frac{1}{T}=\frac{1}{0.5309}$=1.8843 Hz.

The angular frequency, $\omega = 2\pi n$

= 2$\times$ 3.14 $\times$ 1.8843

= 11.833 rad/sec.

(c) For the amplitude, We have

$$v = \omega\sqrt{a^2 - y^2}$$

$$or, 0.4 = 11.833\sqrt{a^2 - (0.03)^2}$$

$$or. (0.4)^2 = (11.833)^2[a^2 - (0.03)^2]$$

$$or, a = 0.04526\, m.$$

Example: A 0.2 kg mass suspended from a spring describes a simple harmonic motion with a period T of 3 s and amplitude R of 10 cm. At t=0 the mass passed upward through the equilibrium position. (a) Find the force constant k of the spring (b) Find the displacement, velocity and acceleration of the mass when t=1 s (c) Also show that at t=1, the sum of the potential and kinetic energy is equal to $\frac{1}{2}ka^2$?

Solution:

(a) $\omega = \sqrt{\frac{k}{m}}$

or, $\omega^2 = \frac{k}{m}$

or, $(2\pi n)^2 = \frac{k}{m}$

$or, \left(\frac{2\pi}{T}\right)^2 = \frac{k}{m}$

$or, \frac{4\pi^2}{T^2} = \frac{k}{m}$

or, k= $\frac{4\pi^2}{3^2} \times 0.2$

=0.88 N/m

(b) Let us consider the upward direction as the positive direction. Then from

$$y = a \sin \omega t$$

At t=1 s,

$$y = 0.10 \sin \left[\frac{2\pi}{3} \times 1\right]$$
$$= 0.10 \sin 120^o$$
$$= 0.0866\text{m}$$

$$v = \frac{dy}{dt} = \omega\, a \cos \omega t$$

$$= \left(\frac{2\pi}{T}\right)(a) \cos\left[\frac{2\pi}{T} \times 1\right]$$

$$= \left(\frac{2\pi}{3}\right)(0.10) cos 120^o$$

= -0.105 m/s

$$\text{acc ln.} = \frac{d^2y}{dt^2} = -\omega^2 a \sin \omega t$$

$$= -\left(\frac{2\pi}{T}\right)^2 (\text{a}) \sin\left[\frac{2\pi}{T} \times 1\right]$$

$$= -\left(\frac{2\pi}{3}\right)^2 (0.10)\sin 120^o$$

$$= -0.38 \text{ m}/s^2$$

(c) E= P.E. + K.E.

$$\frac{1}{2}ka^2 = \frac{1}{2}ky^2 + \frac{1}{2}mv^2$$

$$\left(\frac{1}{2}\right)(0.88)(0.1)^2 = \left(\frac{1}{2}\right)(0.88)(0.0866)^2 + \left(\frac{1}{2}\right)(0.2)(0.105)^2$$

$$4.4 \times 10^{-3} J = (3.3 \times 10^{-3} + 1.1 \times 10^{-3}) J$$

$$= 4.4 \times 10^{-3} J$$

Example: The diaphragm of a loudspeaker moves back and forth in simple harmonic motion to create sound. The frequency of the motion is f= 1.0 kHz and the amplitude is A= 0.20mm

(a) What is the maximum speed of the diaphragm?

(b) Where in the motion does this maximum speed occur?

Solution:

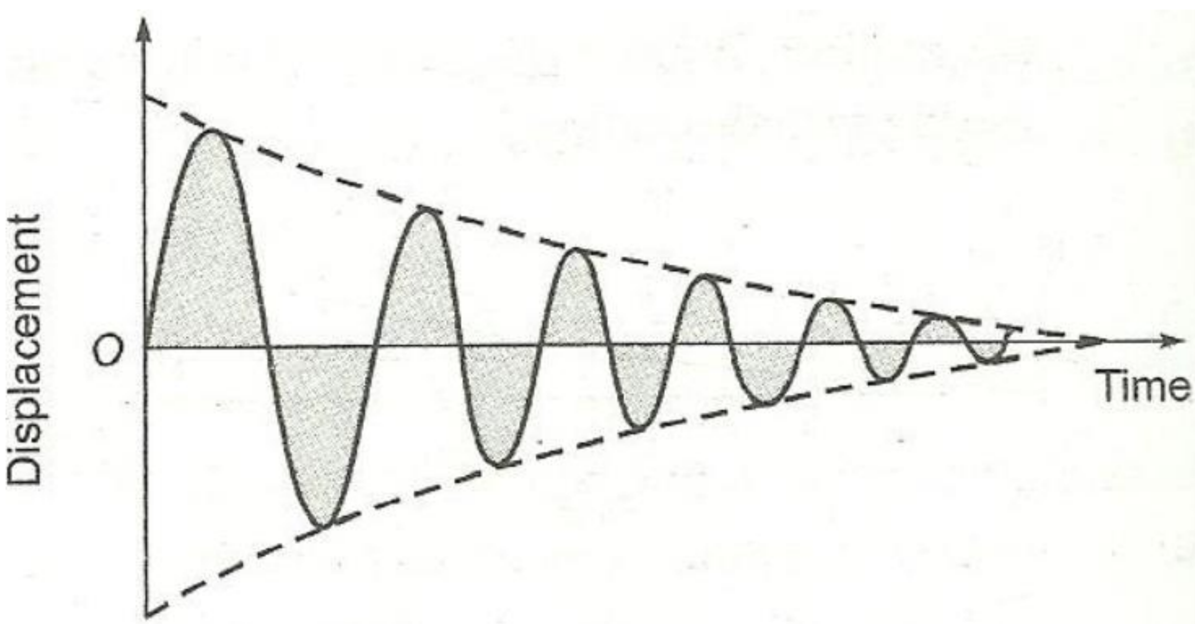
(a) V_{max}= $A\omega = A(2\pi f) = (0.20 \times 10^{-3})2\pi(1.0 \times 10^{3})$

$$= 1.3 m/s$$

(b) The speed of the diaphragm is zero when the diaphragm momentarily comes to rest at either end of its motion: x= +A and x=-A. Its maximum speed occurs midway between these two positions, or at x = 0 m.

Damped Oscillations

Damped Harmonic Motion: When oscillating bodies do not move back and forth between precisely fixed limits because frictional force dissipate the energy and amplitude of oscillation decreases with time and finally die out. Such harmonic motion is called Damped Harmonic Motion.



Damped Oscillation is defined as the reduction in amplitude of an oscillating system due to the dissipation of energy. The rate at which the amplitude decreases is proportional to the square of the amplitude. There are three types of damped oscillation, underdamped, overdamped, and critically damped oscillation.

Thus a body executing simple harmonic oscillations in a damping medium will be simultaneously subjected to the following two opposite forces:

(1) The restoring force acting on the body which is proportional to the displacement of the body and acts in a direction opposite to the displacement. Let this force be –ay where a is the force constant.

(2) A resistive or damping force. It has been shown by Mayevski that at ordinary velocities, the opposing, resistive or damping fortce is, to a first approximation, proportional to the velocity of the oscillating body.

As most cases of interest to us fall in the category of ordinary velocities, the damping or resistive force may thus be represented by

$F = -bv = -b\frac{dy}{dt}$

Where b is a constant of proportionality. b is a positive constant called damping coefficient of the medium and may be looked upon as the resistive or dissipative force per unit velocity. The negative sign signifies a restraining influence on the vibration of the particle.

Thus, the differential equation of motion of a body executing damped harmonic oscillations may be written as

$$m\frac{d^2y}{dt^2} = -ay - b.\frac{dy}{dt}$$

$$\text{or, } m\frac{d^2y}{dt^2} + b.\frac{dy}{dt} + ay = 0$$

or, $\frac{d^2y}{dt^2} + \frac{b}{m}\frac{dy}{dt} + \frac{a}{m}y$=0

or, , $\frac{d^2y}{dt^2} + 2\gamma\frac{dy}{dt} + \omega^2 y$=0

Where $2\gamma = \frac{b}{m}$ and $\omega^2 = \frac{a}{m}$

This equation is referred to as the differential equation of a damped harmonic oscillator.